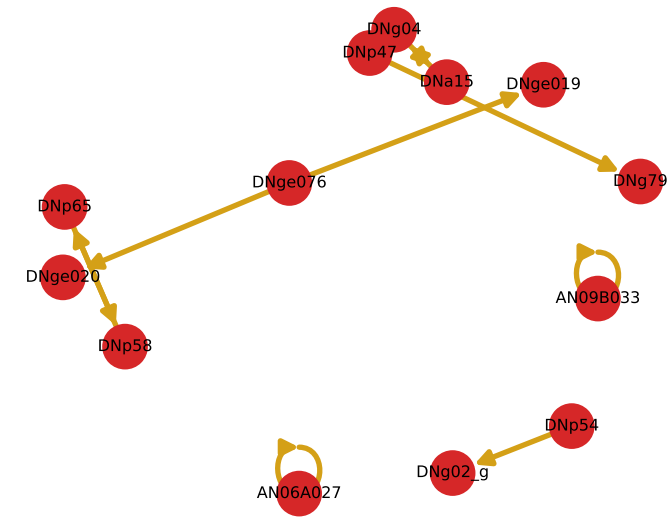


A Conserved Sensorimotor Circuit across Three Drosophila Connectomes

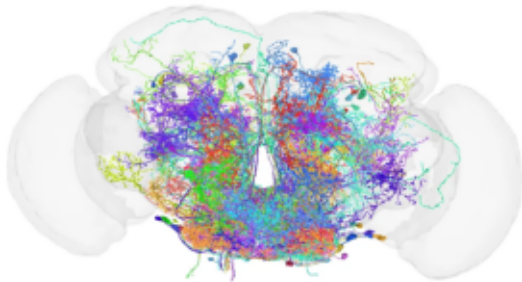
Research summary · dataset: FAFB (♀ adult brain, Dorkenwald et al. 2024) · Yanjin Li

(1) Conserved-circuit network graph



105 neurons · 12 directed edges identical in BANC×FAFB×MANC (gold).

(2) Codex 3D meshes



All 105 circuit neurons rendered as Codex 3D meshes inside the FAFB whole-brain, converging at the midline / cervical connective ([codex.flywire.ai](#); live link in [results/codex_3d_url.txt](#), IDs in [codex_circuit_ids.txt](#)).

(3) Observations & biological hypothesis

- The conserved set is almost entirely sensorimotor: descending 66.2× and ascending 25.0× enriched vs the FAFB whole-brain background (DN+AN ≈ 93% of the circuit), giving direct structural support for the sensorimotor bottleneck / effectome (Pospisil et al. 2024).
- Specific wiring is conserved far beyond degree sequence: 2609 all-three consensus edges vs 38 expected under a degree-preserving null (68.9×, $Z=475\sigma$) — the agreement reflects real connectivity identity, not matched degree.
- 88.6% of the circuit is conserved across sexes (♀ FAFB/BANC vs ♂ MANC); the few dimorphic neurons project to abdominal VNC / lateral brain — the classes expected to diverge for reproductive behaviour.
- The conserved edges form reciprocal DN↔DN pairs (DNa15↔DNg04, DNp58↔DNp65) with mixed ACh/GABA chemistry — a feedforward-inhibition motif (Milo et al. 2002) plausibly filtering descending motor commands.
- Hub neurons span multiple motor modules (wing/flight, leg, abdominal) and include DNg02 — a documented flight-motor population that sets wingbeat amplitude via a population code (Schnell et al. 2022): a known controller sits inside the invariant backbone.
- Hypothesis: these edges are a developmentally canalised brain↔cord communication backbone; silencing the hub neurons should impair walking, flight and posture simultaneously (testable via optogenetics).

(4) References

1. Dorkenwald et al. (2024) Nature 634:123 — FAFB connectome.
2. Schlegel et al. (2024) Nature 634:139 — multi-connectome cell typing / NBLAST.
3. Bates et al. (2025) bioRxiv — BANC; cross-dataset correspondence metadata.
4. Pospisil et al. (2024) Nature 634:234 — sensorimotor bottleneck / effectome.
5. Witvliet et al. (2021) Nature 596:257 — connectome stereotypy.
6. Milo et al. (2002) Science 298:824 — network motifs (feedforward inhibition).